

The Expected Value

- A Random Variable is a set of possible values from a random experiment.
- Example: Tossing a coin: we could get Heads or Tails.
- Let's give them the values Heads=0 and Tails=1 and we have a Random Variable "X":
- random variable 1

Mean or Expected value of Random Variable

Mean, or Expected Value of a random variable X

Let X be a random variable with probability distribution $f(x)$. The mean, or expected value, of X is

$$\mu = E(X) = \begin{cases} \sum_{-\infty}^{\infty} x f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

- Mean or Expected Value: μ
- When we know the probability p of every value x we can calculate the Expected Value (Mean) of X :
- $\mu = \sum xp$
- To calculate the Expected Value:
- multiply each value by its probability
- sum them up

- Example: Tossing a single unfair die
- For fun, imagine a weighted die (cheating!) so we have these probabilities:

• 1	2	3	4	5	6
• 0.1	0.1	0.1	0.1	0.1	0.5

Example :

• <u>x</u>	1	2	3	4	5	6
• <u>p</u>	0.1	0.1	0.1	0.1	0.1	0.5

• xp 0.1 0.2 0.3 0.4 0.5 3

• $\mu = \sum xp = 0.1+0.2+0.3+0.4+0.5+3 = 4.5$

• The expected value is 4.5

Example 8.1. Roll one die. Then $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let X be the value on the die. So, $X(\omega) = \omega$. If the die is fair, then the probability model has $P\{\omega\} = 1/6$ for each outcome ω . Using the formula (8.1), the expected value

$$\begin{aligned} EX &= 1 \cdot P\{1\} + 2 \cdot P\{2\} + 3 \cdot P\{3\} + 4 \cdot P\{4\} + 5 \cdot P\{5\} + 6 \cdot P\{6\} \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{21}{6} = \frac{7}{2}. \end{aligned}$$

An example of an unfair dice would be the probability with $P\{1\} = P\{2\} = P\{3\} = 1/4$ and $P\{4\} = P\{5\} = P\{6\} = 1/12$. In this case, the expected value

$$EX = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{12} + 6 \cdot \frac{1}{12} = \frac{11}{4}.$$

The *variance* of a random variable X , or the variance of the probability distribution of X , is defined as the expected squared deviation from the expected value.

Variance & Standard Deviation

Let X be a random variable with probability distribution $f(x)$ and mean μ . The *variance* of X is

$$\begin{aligned}\sigma^2 &= \text{Var}(X) \\ &= E[(X - \mu)^2] \\ &= E[(X - E(X))^2] \\ &= \begin{cases} \sum (x - \mu)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx & \text{if } X \text{ is continuous} \end{cases}\end{aligned}$$

The positive square root of the variance, σ , is called the standard deviation of X .

EXAMPLE 4.14. Refer to Example 4.1 (Discrete). Find $\sigma^2 = \text{Var}(X)$, the variance of X .

Note that

$$\begin{aligned}E[(X - \mu)^2] &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - \mu^2\end{aligned}$$

We often calculate the variance in the following way:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Show that the variance of a random variable equals the Expected Value of the square of the random variable minus the square of the Expected Value of the random variable.

$$\text{Show } \text{Var}[X] = E[X^2] - (E[X])^2$$

proof: by definition

$$\begin{aligned} \text{Var}[X] &= E[(X - E[X])^2] = E[(X - E[X])(X - E[X])] \\ &= E[X^2 - X E[X] - X E[X] + (E[X])^2] \\ &= E[X^2 - 2X E[X] + (E[X])^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - E[2\mu X] + E[\mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 = E[X^2] - (E[X])^2 \end{aligned}$$

$$E[X] = \mu$$

$$E[k] = k$$

$$E[kX] = k E[X]$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\begin{aligned}\text{Var}(X) &= E[(X - \mu)^2] \\&= \sum_x (x - \mu)^2 p(x) \\&= \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\&= E[X^2] - 2\mu^2 + \mu^2 \\&= E[X^2] - \mu^2\end{aligned}$$

Variance: $\text{Var}(X)$

- The Variance is:
- $\text{Var}(X) = \sum x^2 p - \mu^2$

To calculate the Variance:

- square each value and multiply by its probability
- sum them up and we get $\sum x^2 p$
- then subtract the square of the Expected Value μ^2

Example :

• x	1	2	3	4	5	6
• p	0.1	0.1	0.1	0.1	0.1	0.5

• x^2p 0.1 0.4 0.9 1.6 2.5 18

• $\Sigma x^2p = 0.1+0.4+0.9+1.6+2.5+18 = 23.5$

• $\text{Var}(X) = \Sigma x^2p - \mu^2 = 23.5 - 4.52 = 3.25$

• **The variance is 3.25**

- You plan to open a new McDougals Fried Chicken, and found these stats for similar restaurants:
- Percent Year's Earnings
- 20% \$50,000 Loss
- 30% \$0
- 40% \$50,000 Profit
- 10% \$150,000 Profit
- Using that as probabilities for your new restaurant's profit, what is the Expected Value and Standard Deviation?